```vba
Option Explicit

'=========================================================
' FIXED BUTTONS – NEVER RESIZE / NEVER MOVE
'=========================================================
Sub SetupReportButtons()

    Dim sht As Worksheet
    Set sht = Sheets("Report")

    Dim shp As Shape
    For Each shp In sht.Shapes
        If shp.Name = "btnCity" Or shp.Name = "btnYear" Then shp.Delete
    Next shp

    '================ CITY BUTTON =================
    Set shp = sht.Shapes.AddShape(msoShapeRoundedRectangle, 20, 15, 240, 36)
    With shp
        .Name = "btnCity"
        .TextFrame.Characters.text = "Generate City Report"
        .OnAction = "City_Report"
        .Fill.ForeColor.RGB = RGB(0, 112, 192)
        .TextFrame.Characters.Font.Color = vbWhite
        .TextFrame.Characters.Font.Bold = True
        .Placement = xlFreeFloating   'Fixed size + fixed position
    End With

    '================ YEAR BUTTON =================
    Set shp = sht.Shapes.AddShape(msoShapeRoundedRectangle, 280, 15, 240, 36)
    With shp
        .Name = "btnYear"
        .TextFrame.Characters.text = "Generate Year Report"
        .OnAction = "Year_Report"
        .Fill.ForeColor.RGB = RGB(0, 176, 80)
        .TextFrame.Characters.Font.Color = vbWhite
        .TextFrame.Characters.Font.Bold = True
        .Placement = xlFreeFloating   'Fixed size + fixed position
    End With

End Sub

'=========================================================
' CLEAR REPORT CONTENT (KEEP BUTTONS)
'=========================================================
Sub ClearReportContent()
    Dim rpt As Worksheet
    Set rpt = Sheets("Report")
```

```
    ' Clear only cells, not shapes
    rpt.Cells.Clear

    ' Recreate buttons
    SetupReportButtons
End Sub

'========================================================
' CITY REPORT WITH VALIDATION
'========================================================
Sub City_Report()

    Dim ws As Worksheet, rpt As Worksheet
    Set ws = Sheets("EV_Data")
    Set rpt = Sheets("Report")

    ' Clear report content but keep buttons
    ClearReportContent

    Dim city As String
    city = Trim(InputBox("Enter City Name:" & vbNewLine & vbNewLine & _
                "Note: Enter exact city name as in data"))

    If city = "" Then
        MsgBox "Operation cancelled.", vbInformation
        Exit Sub
    End If

    ' Validate if city exists in data
    If Not CityExists(ws, city) Then
        MsgBox "City '" & city & "' not found in the data!" & vbNewLine & vbNewLine & _
            "Please enter a valid city name.", vbExclamation, "City Not Found"
        Exit Sub
    End If

    GenerateReport ws, rpt, city, 0, "City"

End Sub

'========================================================
' YEAR REPORT WITH VALIDATION
'========================================================
Sub Year_Report()

    Dim ws As Worksheet, rpt As Worksheet
    Set ws = Sheets("EV_Data")
    Set rpt = Sheets("Report")
```

```vba
    ' Clear report content but keep buttons
    ClearReportContent

    Dim yr As Variant
    yr = Trim(InputBox("Enter Year (e.g. 2021):" & vbNewLine & vbNewLine & _
                 "Available years: 2018-2024"))

    If yr = "" Then
        MsgBox "Operation cancelled.", vbInformation
        Exit Sub
    End If

    ' Validate year is numeric
    If Not IsNumeric(yr) Then
        MsgBox "Please enter a valid year (numeric value).", vbExclamation, "Invalid Year"
        Exit Sub
    End If

    ' Validate year range
    If Val(yr) < 2010 Or Val(yr) > 2030 Then
        MsgBox "Please enter a realistic year between 2010 and 2030.", vbExclamation, "Year Out of Range"
        Exit Sub
    End If

    ' Validate if year exists in data
    If Not YearExists(ws, Val(yr)) Then
        MsgBox "Year '" & yr & "' not found in the data!" & vbNewLine & vbNewLine & _
               "Please enter a valid year that exists in the dataset.", vbExclamation, "Year Not Found"
        Exit Sub
    End If

    GenerateReport ws, rpt, "", Val(yr), "Year"

End Sub

'========================================================
' VALIDATION FUNCTIONS
'========================================================
Function CityExists(ws As Worksheet, city As String) As Boolean
    Dim lastRow As Long, i As Long
    lastRow = ws.Cells(ws.Rows.count, "A").End(xlUp).row

    For i = 2 To lastRow
        If LCase(Trim(ws.Cells(i, 1).Value)) = LCase(Trim(city)) Then
            CityExists = True
            Exit Function
        End If
    Next i
```

```vba
    CityExists = False
End Function

Function YearExists(ws As Worksheet, yr As Long) As Boolean
    Dim lastRow As Long, i As Long
    lastRow = ws.Cells(ws.Rows.count, "D").End(xlUp).row

    For i = 2 To lastRow
        If ws.Cells(i, 4).Value = yr Then
            YearExists = True
            Exit Function
        End If
    Next i

    YearExists = False
End Function

'==========================================================
' GET TOTAL EVS FOR A SPECIFIC YEAR
'==========================================================
Function GetYearTotal(ws As Worksheet, yr As Long) As Long
    Dim lastRow As Long, i As Long, count As Long
    lastRow = ws.Cells(ws.Rows.count, "D").End(xlUp).row
    count = 0

    For i = 2 To lastRow
        If ws.Cells(i, 4).Value = yr Then
            count = count + 1
        End If
    Next i

    GetYearTotal = count
End Function

'==========================================================
' GET YOY GROWTH ANALYSIS
'==========================================================
Sub GetYoYGrowth(ws As Worksheet, yr As Long, rpt As Worksheet, startRow As Long)
    ' Get current year total
    Dim currentYearTotal As Long
    currentYearTotal = GetYearTotal(ws, yr)

    ' Get previous year total
    Dim previousYearTotal As Long
    previousYearTotal = GetYearTotal(ws, yr - 1)

    ' Create YoY Growth section
```

```
Dim row As Long
row = startRow

' YoY Growth Header
rpt.Cells(row, 1).Value = "YoY Growth Analysis"
rpt.Cells(row, 1).Font.Bold = True
rpt.Cells(row, 1).Font.Size = 11
rpt.Cells(row, 1).Font.Color = RGB(0, 112, 192) ' Blue color

row = row + 1

' Table Header
rpt.Cells(row, 1).Value = "Year"
rpt.Cells(row, 2).Value = "EV Registrations"
rpt.Cells(row, 3).Value = "YoY Growth %"
rpt.Cells(row, 4).Value = "Trend"

With rpt.Range(rpt.Cells(row, 1), rpt.Cells(row, 4))
  .Font.Bold = True
  .Borders.LineStyle = xlContinuous
  .Borders.Weight = xlThin
  .Interior.Color = RGB(240, 240, 240) ' Light gray
End With

row = row + 1

' Previous Year Data
rpt.Cells(row, 1).Value = yr - 1
rpt.Cells(row, 2).Value = previousYearTotal
rpt.Cells(row, 3).Value = "-"
rpt.Cells(row, 4).Value = "Base Year"

With rpt.Range(rpt.Cells(row, 1), rpt.Cells(row, 4)).Borders
  .LineStyle = xlContinuous
  .Weight = xlThin
End With

row = row + 1

' Current Year Data
rpt.Cells(row, 1).Value = yr
rpt.Cells(row, 2).Value = currentYearTotal

' Calculate YoY Growth
Dim yoyGrowth As Double
If previousYearTotal > 0 Then
  yoyGrowth = ((currentYearTotal - previousYearTotal) / previousYearTotal) * 100
  rpt.Cells(row, 3).Value = Round(yoyGrowth, 1) & "%"
```

```
    ' Add trend indicator
    If yoyGrowth > 0 Then
        rpt.Cells(row, 4).Value = "? Growth"
        rpt.Cells(row, 4).Font.Color = RGB(0, 176, 80) ' Green for growth
        rpt.Cells(row, 4).Font.Bold = True
    ElseIf yoyGrowth < 0 Then
        rpt.Cells(row, 4).Value = "? Decline"
        rpt.Cells(row, 4).Font.Color = RGB(255, 0, 0) ' Red for decline
        rpt.Cells(row, 4).Font.Bold = True
    Else
        rpt.Cells(row, 4).Value = "No Change"
    End If
Else
    rpt.Cells(row, 3).Value = "N/A"
    rpt.Cells(row, 4).Value = "No previous data"
End If

With rpt.Range(rpt.Cells(row, 1), rpt.Cells(row, 4)).Borders
    .LineStyle = xlContinuous
    .Weight = xlThin
End With

' Highlight current year row
With rpt.Range(rpt.Cells(row, 1), rpt.Cells(row, 4)).Interior
    .Color = RGB(255, 255, 204) ' Light yellow highlight
End With

row = row + 2

' Add Growth Insight
If previousYearTotal > 0 Then
    If yoyGrowth > 50 Then
        rpt.Cells(row, 1).Value = "?? Explosive Growth: " & Round(yoyGrowth, 1) & "% YoY increase!"
        rpt.Cells(row, 1).Font.Color = RGB(0, 128, 0) ' Dark green
    ElseIf yoyGrowth > 20 Then
        rpt.Cells(row, 1).Value = "?? Strong Growth: " & Round(yoyGrowth, 1) & "% YoY increase"
        rpt.Cells(row, 1).Font.Color = RGB(0, 176, 80) ' Green
    ElseIf yoyGrowth > 0 Then
        rpt.Cells(row, 1).Value = "?? Moderate Growth: " & Round(yoyGrowth, 1) & "% YoY increase"
        rpt.Cells(row, 1).Font.Color = RGB(146, 208, 80) ' Light green
    ElseIf yoyGrowth < 0 Then
        rpt.Cells(row, 1).Value = "?? Market Decline: " & Round(Abs(yoyGrowth), 1) & "% YoY decrease"
        rpt.Cells(row, 1).Font.Color = RGB(255, 0, 0) ' Red
    Else
        rpt.Cells(row, 1).Value = "?? Stable Market: No YoY change"
        rpt.Cells(row, 1).Font.Color = RGB(255, 192, 0) ' Orange
    End If
```

```
        rpt.Cells(row, 1).Font.Bold = True
    End If

    ' Auto-fit columns
    rpt.Columns(1).AutoFit
    rpt.Columns(2).AutoFit
    rpt.Columns(3).AutoFit
    rpt.Columns(4).AutoFit

End Sub

'=========================================================
' REPORT CORE WITH DATA-DRIVEN INSIGHTS
'=========================================================
Sub GenerateReport(ws As Worksheet, rpt As Worksheet, city As String, yr As Long, reportType As String)

    Dim dMake As Object, dModel As Object, dEV As Object, dCAFV As Object, dUtil As Object
    Set dMake = CreateObject("Scripting.Dictionary")
    Set dModel = CreateObject("Scripting.Dictionary")
    Set dEV = CreateObject("Scripting.Dictionary")
    Set dCAFV = CreateObject("Scripting.Dictionary")
    Set dUtil = CreateObject("Scripting.Dictionary")

    Dim i As Long, lastRow As Long, totalEVs As Long
    lastRow = ws.Cells(ws.Rows.count, "A").End(xlUp).row

    For i = 2 To lastRow

        If city <> "" And LCase(Trim(ws.Cells(i, 1).Value)) <> LCase(Trim(city)) Then GoTo SkipRow
        If yr <> 0 And ws.Cells(i, 4).Value <> yr Then GoTo SkipRow

        totalEVs = totalEVs + 1
        AddCount dMake, ws.Cells(i, 5).Value
        AddCount dModel, ws.Cells(i, 6).Value
        AddCount dEV, ws.Cells(i, 7).Value
        AddCount dCAFV, ws.Cells(i, 8).Value
        AddCount dUtil, ws.Cells(i, 13).Value

SkipRow:
    Next i

    ' Check if any data was found
    If totalEVs = 0 Then
        If reportType = "City" Then
            MsgBox "No data found for city: " & city, vbExclamation, "No Data"
        Else
            MsgBox "No data found for year: " & yr, vbExclamation, "No Data"
        End If
```

```
        Exit Sub
    End If

    '================ MAIN HEADER =================
    rpt.Range("A5").Value = IIf(city <> "", "CITY REPORT – " & city, "YEAR REPORT – " & yr)
    rpt.Range("A5").Font.Size = 14
    rpt.Range("A5").Font.Bold = True

    rpt.Range("A6").Value = "Total EVs:"
    rpt.Range("B6").Value = totalEVs
    rpt.Range("B6").Font.Bold = True

    '================ GENERATE INSIGHTS FROM DATA =================
    GenerateInsights rpt, dMake, dModel, dEV, dCAFV, dUtil, totalEVs, reportType, city, yr

    '================ ADD YOY GROWTH FOR YEAR REPORTS =================
    Dim growthStartRow As Long
    growthStartRow = 15  ' Start after insights

    If reportType = "Year" Then
        GetYoYGrowth ws, yr, rpt, growthStartRow
        growthStartRow = growthStartRow + 12  ' Adjust for YoY growth section
    End If

    '================ DATA SECTIONS - COMPACT LAYOUT =================
    Dim topMakesEndRow As Long
    Dim topModelsEndRow As Long
    Dim utilitiesEndRow As Long

    ' ROW 1: Top Makes, Top Models, Utilities (side by side)
    topMakesEndRow = OutputSectionCompact(rpt, "Top Makes", dMake, growthStartRow, 1)
    topModelsEndRow = OutputSectionCompact(rpt, "Top Models", dModel, growthStartRow, 4)
    utilitiesEndRow = OutputSectionCompact(rpt, "Electric Utility", dUtil, growthStartRow, 7)

    ' Find the maximum end row from first row
    Dim maxFirstRowEnd As Long
    maxFirstRowEnd = Application.Max(topMakesEndRow, topModelsEndRow, utilitiesEndRow)

    ' Add only 2 rows gap
    Dim secondRowStart As Long
    secondRowStart = maxFirstRowEnd + 2

    ' ROW 2: EV Types and CAFV Eligibility
    OutputSectionCompact rpt, "EV Types", dEV, secondRowStart, 1
    OutputSectionCompact rpt, "CAFV Eligibility", dCAFV, secondRowStart, 4

End Sub
```

```vba
'====================================================
' GENERATE DATA-DRIVEN INSIGHTS
'====================================================
Sub GenerateInsights(rpt As Worksheet, dMake As Object, dModel As Object, dEV As Object, dCAFV As Object, dUtil As Object, totalEVs As Long, reportType As String, city As String, yr As Long)

    Dim insightRow As Long
    insightRow = 8

    rpt.Cells(insightRow, 1).Value = "Data Insights:"
    rpt.Cells(insightRow, 1).Font.Bold = True
    rpt.Cells(insightRow, 1).Font.Size = 11

    insightRow = insightRow + 1

    ' Get top values
    Dim makeKeys, makeVals, evKeys, evVals, cafvKeys, cafvVals, utilKeys, utilVals
    makeKeys = dMake.keys: makeVals = dMake.Items
    evKeys = dEV.keys: evVals = dEV.Items
    cafvKeys = dCAFV.keys: cafvVals = dCAFV.Items
    utilKeys = dUtil.keys: utilVals = dUtil.Items

    SortDesc makeKeys, makeVals
    SortDesc evKeys, evVals
    SortDesc cafvKeys, cafvVals
    SortDesc utilKeys, utilVals

    Dim insightsCount As Long
    insightsCount = 0

    ' Insight 1: Top Make dominance
    If dMake.count > 0 And totalEVs > 0 Then
        Dim topMakeShare As Double
        topMakeShare = Round(makeVals(0) / totalEVs * 100, 1)

        If reportType = "City" Then
            rpt.Cells(insightRow, 1).Value = "• " & makeKeys(0) & " dominates with " & topMakeShare & "% of all EVs in " & city
        Else
            rpt.Cells(insightRow, 1).Value = "• " & makeKeys(0) & " leads in " & yr & " with " & topMakeShare & "% market share"
        End If
        insightRow = insightRow + 1
        insightsCount = insightsCount + 1
    End If

    ' Insight 2: EV Type analysis
    If dEV.count > 0 And totalEVs > 0 Then
        Dim bevCount As Long, phevCount As Long, totalBevPhev As Long
```

```
        bevCount = 0: phevCount = 0

        Dim evKey
        For Each evKey In dEV.keys
            If InStr(1, UCase(evKey), "BEV") > 0 Or InStr(1, UCase(evKey), "BATTERY") > 0 Then
                bevCount = bevCount + dEV(evKey)
            ElseIf InStr(1, UCase(evKey), "PHEV") > 0 Or InStr(1, UCase(evKey), "PLUG-IN") > 0 Then
                phevCount = phevCount + dEV(evKey)
            End If
        Next evKey

        totalBevPhev = bevCount + phevCount
        If totalBevPhev > 0 Then
            Dim bevShare As Double, phevShare As Double
            bevShare = Round(bevCount / totalBevPhev * 100, 1)
            phevShare = Round(phevCount / totalBevPhev * 100, 1)

            rpt.Cells(insightRow, 1).Value = "• BEVs account for " & bevShare & "% vs PHEVs at " & phevShare
& "%"
            insightRow = insightRow + 1
            insightsCount = insightsCount + 1
        End If
    End If

    ' Insight 3: CAFV eligibility
    If dCAFV.count > 0 And totalEVs > 0 Then
        Dim eligibleCount As Long, nonEligibleCount As Long
        eligibleCount = 0: nonEligibleCount = 0

        Dim cafvKey
        For Each cafvKey In dCAFV.keys
            If InStr(1, UCase(cafvKey), "ELIGIBLE") > 0 Or InStr(1, UCase(cafvKey), "YES") > 0 Then
                eligibleCount = eligibleCount + dCAFV(cafvKey)
            ElseIf InStr(1, UCase(cafvKey), "NOT") > 0 Or InStr(1, UCase(cafvKey), "NO") > 0 Then
                nonEligibleCount = nonEligibleCount + dCAFV(cafvKey)
            End If
        Next cafvKey

        Dim totalCafv As Long
        totalCafv = eligibleCount + nonEligibleCount
        If totalCafv > 0 Then
            Dim eligibleShare As Double
            eligibleShare = Round(eligibleCount / totalCafv * 100, 1)

            rpt.Cells(insightRow, 1).Value = "• " & eligibleShare & "% of EVs are CAFV eligible"
            insightRow = insightRow + 1
            insightsCount = insightsCount + 1
        End If
```

```
    End If

    ' Insight 4: Top Utility provider
    If dUtil.count > 0 And totalEVs > 0 And utilKeys(0) <> "(Blank)" Then
        Dim topUtilShare As Double
        topUtilShare = Round(utilVals(0) / totalEVs * 100, 1)

        If reportType = "City" Then
            rpt.Cells(insightRow, 1).Value = "• " & utilKeys(0) & " serves " & topUtilShare & "% of EVs in " & city
        Else
            rpt.Cells(insightRow, 1).Value = "• " & utilKeys(0) & " is the main utility provider (" & topUtilShare
& "%)"
        End If
        insightRow = insightRow + 1
        insightsCount = insightsCount + 1
    End If

    ' Insight 5: Make diversity
    If dMake.count > 1 And totalEVs > 0 Then
        Dim top3Share As Double
        top3Share = 0
        Dim j As Long
        For j = 0 To Application.Min(2, dMake.count - 1)
            top3Share = top3Share + makeVals(j)
        Next j
        top3Share = Round(top3Share / totalEVs * 100, 1)

        rpt.Cells(insightRow, 1).Value = "• Top 3 makes represent " & top3Share & "% of total EVs"
        insightRow = insightRow + 1
        insightsCount = insightsCount + 1
    End If

    ' If no insights were generated (edge case)
    If insightsCount = 0 Then
        rpt.Cells(insightRow, 1).Value = "• Data analysis completed successfully"
    End If

    ' Add only 1 blank row after insights
    insightRow = insightRow + 1

End Sub

'=======================================================
' OUTPUT SECTION - COMPACT VERSION
'=======================================================
Function OutputSectionCompact(rpt As Worksheet, title As String, d As Object, r As Long, c As Long) As
Long
    ' Returns the ending row of this section
```

```
    rpt.Cells(r, c).Value = title
    rpt.Cells(r, c).Font.Bold = True

    rpt.Cells(r + 1, c).Value = "Item"
    rpt.Cells(r + 1, c + 1).Value = "Count"
    rpt.Range(rpt.Cells(r + 1, c), rpt.Cells(r + 1, c + 1)).Font.Bold = True

    ' Add borders to header
    With rpt.Range(rpt.Cells(r + 1, c), rpt.Cells(r + 1, c + 1)).Borders
        .LineStyle = xlContinuous
        .Weight = xlThin
    End With

    Dim k, v, i As Long
    k = d.keys: v = d.Items
    SortDesc k, v

    Dim itemsToShow As Long
    itemsToShow = Application.Min(6, d.count) ' Show max 6 items

    For i = 0 To itemsToShow - 1
        rpt.Cells(r + 2 + i, c).Value = k(i)
        rpt.Cells(r + 2 + i, c + 1).Value = v(i)

        ' Add borders to data
        With rpt.Range(rpt.Cells(r + 2 + i, c), rpt.Cells(r + 2 + i, c + 1)).Borders
            .LineStyle = xlContinuous
            .Weight = xlThin
        End With
    Next i

    ' Auto-fit columns for better visibility
    rpt.Columns(c).AutoFit
    rpt.Columns(c + 1).AutoFit

    ' Return ending row
    OutputSectionCompact = r + 2 + itemsToShow - 1

End Function

'==========================================================
' HELPERS
'==========================================================
Sub AddCount(d As Object, k As String)
    If k = "" Then k = "(Blank)"
    If d.Exists(k) Then d(k) = d(k) + 1 Else d.Add k, 1
End Sub
```

```
Sub SortDesc(keys, vals)
    Dim i As Long, j As Long, t1, t2
    For i = LBound(vals) To UBound(vals)
        For j = i + 1 To UBound(vals)
            If vals(j) > vals(i) Then
                t1 = vals(i): vals(i) = vals(j): vals(j) = t1
                t2 = keys(i): keys(i) = keys(j): keys(j) = t2
            End If
        Next j
    Next i
End Sub
```